

classless inter-domain routing (CIDR) addressing are provided in IPv6.

- Provision of extension header in IPv6 meets the requirements of such things as checksum, security options, etc.

- Flow-level and priority headers are used to support real-time services. By assigning higher priority to real-time packets, necessity of time sensitiveness is restored. Data packets and, for that purpose, time-insensitive packets are assigned low priority and serviced by the best effort approach.

- IPv6 ensures support for real-time services.

- IPv6 provides security support that could eventually be seen as its biggest advantage. Today, billion-dollar businesses are carried out over the Internet. To keep businesses secure, a public crypto system has emerged as one of the most important tools. IPv6 with its ancillary security protocol has provided a better communication tool for transacting business activities over the Internet.

- IPv6 has the provision for enhanced routing capability, including support for mobile hosts.

Functions of IPv6 headers bytes are given below.

- Version field (4 bits) contains the version number of Internet Protocol. As of now, there are only two versions: 4 and 6. For version 6, this field is six, which makes four bits of this field as 0110.

- Priority (4 bits) field is used to indicate the priority of the datagram. Four bits in the field define $2^4 (= 16)$ priority levels. The first eight priority levels (from 0000 to 0111) are assigned for the services requiring congestion control. If congestion occurs, traffic is backed off.

These are suitable for conventional data and non-real-time services. Different priority levels under the first eight levels are used as 0 for no priority, 1 for background traffic like net-news, 2 for unattended transfer like e-mails, 3 remains reserved, 4 for attended bulk transfer like File Transfer Protocol (FTP), 5 remains reserved, 6 for interactive traffic such as Telnet, X-windows and so on, and 7 for control traffic such as Simple Network Management Protocol (SNMP) and

routing protocols.

Higher priority levels (from eight to fifteen) are used for services that do not back off in response to congestion such as real-time traffic. Priority levels eight and fifteen are, respectively, for traffic that is most willing to be discarded on congestion and for traffic that is least willing to be discarded on congestion.

- Flow level (24 bits) is used to identify different data flow characteristics, which may be assigned by the source and can be used to label packets. Packet labels may be required to provide special handling of packets by IPv6 routers. One example of flow-level character is QoS or real-time services. The combination of sender IP address and flow label creates a unique path identifier that is used to route datagrams more efficiently.

Each flow is uniquely defined by the combination of the source address and a non-zero flow label. Flow label can be from 000001_h to $FFFFFF_h$ in hex. Packets having no flow label are given a zero label. All packets in the same flow must have the same flow label, same source and destination addresses, and same priority level. The initial flow label is obtained by the source using a pseudo random generator, and subsequent flow numbers are obtained sequentially.

- Payload length (16 bits) field indicates the total size of the payload in bytes of the IP datagram that excludes header fields.

- Next header (8 bits) field indicates the header that follows the IP header. The next header can either be one of the optional extension headers used by IP or the header for an upper layer protocols such as UDP or TCP. The field defines the type of extension header. For example, 0, 1, 6, 44, 51 and 80, respectively are for IP, Internet Control Message Protocol (ICMP), TCP, fragmentation header, authentication header and International Standard Organization (ISO).

Each extension header is made of two fields: extension header field and header length field. When there is no need for any other extension header, the next header will be TCP and, hence, it will contain 6. (Here, 6 indicates that there are no more extension

headers. And, 6 is written in binary as 0110 in the header.) Length of the base header is fixed at 40 bytes. Extension header gives functional flexibility to IPv6 datagram. Maximum six extension headers can be used.

Extension headers are used for source routing, fragmentation, authentication, security, etc. IPv6 currently defines six extension headers: hop-by-hop option header, routing header, fragment header, authentication header, encrypted security payload header and destination options header. If one or more extension headers are used, these must be in the order as stated above. For example, if destination option, authentication and hop-by-hop option headers are to be used, the extension header fields must be: main IPv6 header, hop-by-hop option header, authentication header, destination option header and then TCP header with data.

Each extension header is associated with one 8-bit next header field. In fragment header, flags and offset fields are fixed at 16 bits. Thus, except in fragment header, in all other extension headers, the next header field is immediately followed by 8-bit extension header length. Header length indicates the length of current extension header in multiple of 8 bytes.

In the last extension header, the next header field contains the value 59. By specifying the next header field as TCP, last of the extension header can be indicated. This is because IP does not only carry TCP segment, it carries UDP pack, ICMP pack, etc, too. To standardise, the last field with value 59 is recommended.

- Hop limit (8 bits) field is used to limit the maximum number of hops that a datagram is allowed to move in the network before it reaches its destination. If after moving the maximum number of hops, as assigned by this field, the datagram does not reach the destination, it is discarded from the network. The field is used to avoid congestion that may be caused by the datagram by repetitive and recursive movements in the network. Each router reduces the hop limit by one before releasing the datagram to the network. When the hop limit reaches zero, it is deleted by the detecting